Response Letter with Annotations

Citation (APA format, using Zotero or other bibliography software)

Schneider, P. J., & Rizoiu, M. A. (2023). The effectiveness of moderating harmful online content. *Proceedings of the National Academy of Sciences*, *120*(34), e2307360120.

Annotations

1. **Purposes**
   - **a.** Motivation (one-sentence summary)

The motivation of this study is to examine the effectiveness of moderating harmful content on social media platforms, specifically the required response time, using models based on self-exciting point processes (Hawkes processes).

   - **b.** Specific Rationale (several bullet points)

- The Digital Services Act (DSA) introduced by the European Union mandated platforms to remove harmful content quickly, within 24 hours, after it is flagged.
- Online harmful content is harmful as it spreads at high speeds and has short lifespans. A harmful post will potentially generate other harmful posts offspring.
- This study used a dataset collected from Twitter. According to previous studies, Twitter has the shortest half-life (24 min), which indicates that harmful posts on Twitter spreads and generates offspring posts more quickly.

   - **c.** Research Questions and/or Hypotheses

- RQ: how to quantify the likely harm caused by harmful content and how to determine the response time for effective mitigation?
  - (1)What is the maximum reaction time required to achieve a given harm reduction(e.g., 20% in Figure 2A)
  - (2)What is the expected harm reduction when content moderation is performed within 24h?

      - Identify IV, DV, mediator, moderator

- Variables
  - (n^*)potential harm: expected number of harmful direct offspring
  - $\tau_{1/2}$ content half-life: time required to generate half of the direct offspring
  - n^*_Δ actual harm: the number of direct harmful reactions a content generates prior to moderation at timeΔ(the number of harmful offspring generated right before moderation)
  - χ harm reduction: percentage of direct and indirect harmful offspring avoided by content moderation
  - (Δ)reaction time- mandated time to remove flagged harmful content
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- Note: The relationships among these variables change depending on the research question. The relation between χ and Δ is the focus of this study. For example, harm reduction and reaction time are estimated within the model.

2. **Method**
    a. Research Design
        - Experiment: conditions, manipulations, etc.
        - Survey: cross-sectional, longitudinal, etc.
        - Content analysis: data source, etc.
        - …
    - The study models the spread of harmful content using Hawkes processes and calculates measures based on the dataset collected on Twitter under two topics.

    b. Dataset
    - Twitter posts collected between 1 July and 31 December 2022, using two hashtags #climatescam (479,051 posts), and #americafirst or #americansfirst (278,899 posts)
    - The author explained the selection of these two hashtags in the supplementary material. Both topics include harmful and non-harmful posts. #amricafirst# has been extensively researched in previous studies and has been found to spread through contagion-like dynamics. #climatescam# is a relevantly new hashtag which succeeds #climatehoax.

    c. The models and equations used are illustrated in the Materials and Methods section in the paper.

3. **Results and Conclusions**
    a. Answers to RQs and Hypotheses
- RQ (1): What is the maximum reaction time required to achieve a given harm reduction ($\chi = 20\%$)
    - Result:
        - due to the significantly higher virality for #climatescam, more time (Δ = 24h) is available for content moderation compared to the nationalism topic (Δ = 2.22h).
        - From the Figure 2A, harmful posts under **#climatescam** tend to have higher potential harm—meaning they are more viral and generate more harmful offspring—so the reaction time required to achieve a 20% harm reduction (i.e., preventing 20% of offspring) is also longer compared to the other topic.
        - The reaction time increases with both half-life ($\tau 1/2$) and the potential harm ($n*$).

- RQ(2): What is the expected harm reduction when content moderation is performed within 24h?
    - Results:
        - The harm reduction at the centroids is $\chi = 29.18\%$ and $\chi = 13.29\%$ for #climatescam and #americafirst.
        - The harm reduction χ increases with the potential harm n*

b. Include Key Figures/Tables

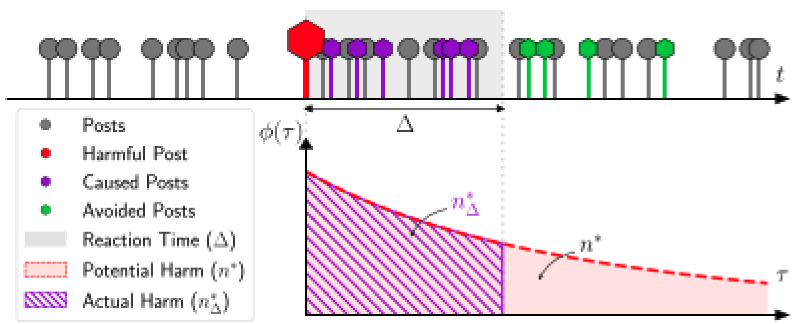

Fig. 1. Social media dynamics as self-exciting point process. Social media posts (●) include a percentage of posts considered harmful (○). One harmful post (●) likely generates n^* other harmful content at the rate $\phi(\tau)$. When the post is removed at time Δ, the likely caused harm is limited to n^*_Δ (●) and further harm is avoided (●). The harm reduction χ is the percentage of all harmful content generated directly or indirectly via self-excitation avoided via moderation.

- This figure illustrates how a self-exciting point process models the spread of harmful content on social media platforms. A harmful post can generate many harmful offspring posts, and the total number of new posts it produces is referred to as its *potential harm*. When the post is removed after a certain reaction time, it prevents the creation of future offspring posts. These avoided posts are counted as the *reduction of harm*.

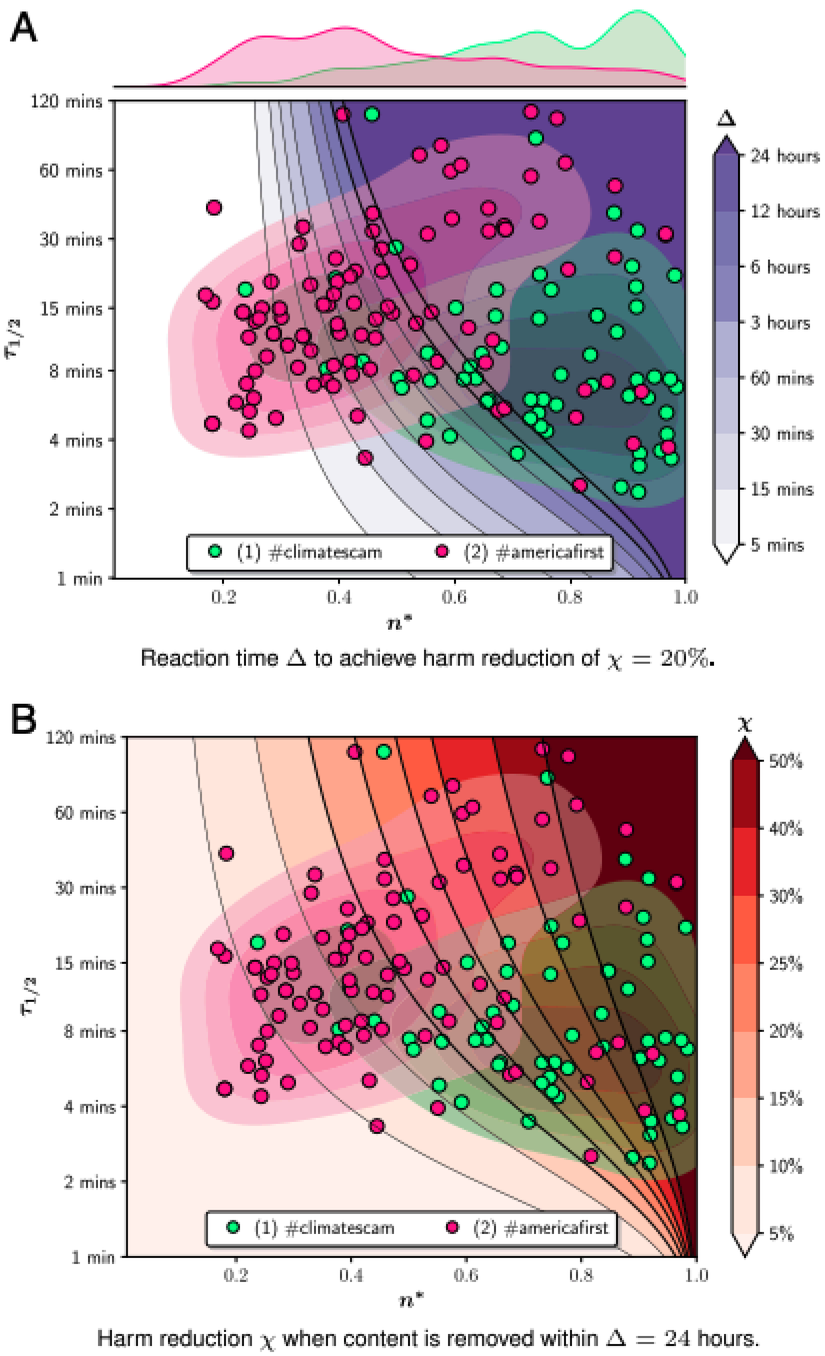

Fig. 2. Visual representation of the dependency of the reaction time Δ (*A*) and harm reduction χ (*B*) with the potential harm n^* (*x*-axis) and content half-live $\tau_{1/2}$ (*y*-axis). Both Δ and χ increase for longer half-lives and higher virality. Real-world potentially problematic content exhibits widely highly variable dynamics. For #climatescam, we can achieve harm reduction of [15%, 50%] for $\Delta = 24$ h.

- From the figure, we can see that posts with the hashtag #climatescam are more harmful, with higher potential harm (i.e., more offspring posts generated). Panel A shows that when we set harm reduction at 20%, the reaction time to achieve this goal is longer for posts with higher potential harm. Within a 24-hour window, most posts in the dataset can reach the goal of reducing harm by 20%. Panel B shows that when the reaction time is fixed at 24 hours, the amount of harm reduction achieved varies across posts with different levels of potential harm and half-life. It indicates that when the moderation time is 24 hours, removing posts with higher potential harm tends to achieve a higher percentage of harm reduction.

c. Authors' Conclusions

- The reaction time increases with both half-life and the potential harm… the latter is significant as it indicates that the DSA-legislated moderation can be effective even on Twitter.
- Harm reduction increases with the potential harm…It indicates that DSA moderation can effectively stop the most harmful content.
- …most harmful content can emerge in both topics…direct the manual flagging efforts toward the most harmful content ($n* > 0.8$) that can be effectively moderated ($\chi > 50\%$)

4. Discussion

a. Implications

- The DSA's requirement of a 24 hour window appears effective to reduce harm for the most harmful content, even on fast-paced platforms like Twitter.
- This paper provides a framework for policymakers to design more effective content moderation strategies and models.
    - Trusted flaggers, mechanisms for reporting harmful content, reaction time for moderation are all important.
    - The flagging and moderation efforts should be directly targeted toward the most harmful content to achieve higher effectiveness.

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Response Letter (~600 words)

- Your assessment of the study
- Weaknesses
- Other notes

This study investigates the effectiveness of harmful content moderation, specifically focusing on the interplay between harm reduction and reaction time, using models based on self-exciting point processes (Hawkes processes). The results suggest that 24 hours is an effective time window to achieve a reduction of harm of 20% for most harmful posts in the dataset. It is most effective for harmful posts with higher potential harm (i.e., the most harmful posts) and can achieve 50% harm reduction.

The visualization in this study is very important for displaying the findings, since the authors did not include many statistical results, and no data-intensive tables are provided in the article. Figure 1 demonstrates the self-exciting property of content spreading on social media platforms. It shows in a vivid way how one harmful post may directly lead to many other harmful posts. The plot with different areas also makes it clear what the key variables denote, including potential harm and actual harm, which are separated by the reaction time (the time of moderation as an intervention). This figure is easy to understand for readers, especially those who do not understand the complicated Hawkes process model and equations in the Materials and Methods section.

The key findings of this study are illustrated by Figure 2, which consists of two panels. The x-axis refers to potential harm (the total number of offspring harmful posts generated without moderation), and the y-axis refers to the content half-life, which means the time required to generate half of the direct offspring posts. The positions of the points in both panels are the same, corresponding to the measures of identified posts from two hashtags in the Twitter dataset. Panel A shows how the reaction time varies for different data points in order to achieve the goal of 20% harm reduction. It is clearly observed that the color becomes darker toward the upper right, indicating longer intervention time. Panel B shows the differing harm reduction percentages when the moderation window is set to 24 hours, which shows a similar pattern to Panel A: harm reduction increases with both half-life and potential harm.

I appreciate the visualizations in this study because the model and equation parts are very challenging for me to understand. Concerning the weakness of the methodology, I do not identify any because I did not get much from the paper. Perhaps due to the word count limit imposed by PNS, this paper is quite concise in its methods and materials, such as sampling and data processing, and does not provide much detail in the supplementary materials. That said, they do provide all the code and data collected from Twitter on GitHub, which could benefit transparency and reproducibility.

Another thought I have is about operationalizing moderation effectiveness as harm reduction, which is defined as the percentage of harmful offspring posts avoided by content moderation. I wonder whether the number of posts avoided is fully equivalent to harm reduction or the effectiveness of moderation. Although this study found that for harmful posts with higher virality (spreading more quickly and leading to more subsequent harmful posts), moderation is more effective and can achieve a higher harm reduction percentage, it is also likely that the remaining harmful content is larger in amount. In addition to the number of posts, other factors, such as the degree of harmfulness and the reach of the audience, can also impact the assessment of content moderation.